

🚀 ML CI/CD Pipeline Architecture on AWS

GitHub Platform

SOURCE CONTROL



GitHub Repository
Code & Configuration

- ML Models (Python)
- Training Scripts
- Dockerfile
- Pipeline Configuration
- Requirements.txt



GitHub Secrets
AWS Credentials

- AWS_ACCESS_KEY_ID
- AWS_SECRET_ACCESS_KEY
- AWS_REGION
- ECR_REPOSITORY
- S3_BUCKET_NAME



TRIGGER ON PUSH/PR

GitHub Actions CI/CD Pipeline

AUTOMATION



Docker Build & Publish
Container Image Pipeline

- Step 1: Checkout code
- Step 2: Build Docker image
- Step 3: Tag with version
- Step 4: Login to AWS ECR
- Step 5: Push to registry



SageMaker Pipeline Runner
ML Workflow Orchestration

- Step 1: Upload CSV data to S3
- Step 2: Configure AWS credentials
- Step 3: Execute Python script
- Step 4: Trigger SageMaker pipeline
- Step 5: Monitor execution



CSV Files (Local)
Training Data

- Raw dataset files
- Feature data
- Labels/targets
- Validation sets



DEPLOY TO AWS CLOUD

AWS Cloud Infrastructure

PRODUCTION



Amazon ECR
Elastic Container Registry

- Docker image storage
- ml-processing:latest
- ml-training:v1.0
- Versioned images
- Vulnerability scanning



Amazon S3 (Data)
Data Lake Storage

- raw/ - Original CSV files
- processed/ - Cleaned data
- features/ - Engineered features
- validation/ - Test sets



EXECUTE ML PIPELINE



SageMaker Processing Job
Data Preprocessing

- Script: data_prep.py
- Data cleaning & validation
- Feature engineering
- Data splitting (train/test)
- Normalization/scaling
- Instance: ml.m5.xlarge



SageMaker Training Job
Model Training

- Script: train_model.py
- Model training (XGBoost/TF/PyTorch)
- Hyperparameter tuning
- Cross-validation
- Model evaluation
- Instance: ml.p3.2xlarge



SAVE RESULTS



Amazon S3 (Models)
Model Artifacts Storage

- models/ - Trained models (.pkl, .h5)
- metrics/ - Performance reports
- logs/ - Training logs
- checkpoints/ - Model versions



CloudWatch
Monitoring & Logging

- Logs: Pipeline execution logs
- Metrics: CPU, Memory, GPU usage
- Alarms: Failure notifications
- Dashboards: Real-time monitoring



Amazon SNS
Notifications

- Pipeline success/failure alerts
- Email notifications
- Slack integration
- Training completion updates

Trigger & Authentication

Process Flow

Data Transfer

Monitoring & Logs